

# YIHANG QIN

Shanghai Jiao Tong University, Shanghai, China

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🎓 [Google Scholar](#)

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## Education

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| <b>M.Phil., Shanghai Jiao Tong University   GPA: 3.63/4.0</b>                  | <b>2024.09 – Present</b> |
| Relevant Course: Artificial Neural Network, Graphs and Networks, Data Mining   | <i>Shanghai, China</i>   |
| <b>B.Eng., Harbin Institute of Technology (Shenzhen)   GPA: 3.33/4.0</b>       | <b>2020.09 – 2024.06</b> |
| Relevant Course: Advanced Mathematics, Automatic Control, Language Programming | <i>Shenzhen, China</i>   |

## Research Interest

Cooperation, Evolutionary Game, Reinforcement Learning, Human-AI Interaction

## Publications

**Yihang Qin** (2026). Exploring cooperation mechanisms via reinforcement learning in network common-pool resource games. Manuscript in preparation.

**Yihang Qin**, H. Chen, & L. Wang. (2026). Recovering cooperation in Spatial Prisoner's Dilemma via a forced loner mechanism. *Physica A: Statistical Mechanics and its Applications*, 131309. doi: 10.1016/j.physa.2026.131309.

**Yihang Qin**, H. Chen, & L. Wang. (2026). A tripartite intervention–conflict game with Frequency-Adjusted Q-learning. *Applied Mathematical Modelling*, 117072. doi: 10.1016/j.apm.2026.117072.

**Yihang Qin**, H. Chen, & L. Wang. (2025, July). An Tripartite Evolutionary Game Analysis of the ‘Siege Wei to Rescue Zhao’ Strategy. In *2025 44th Chinese Control Conference (CCC)* (pp. 8458–8463). IEEE. doi: 10.23919/CCC64809.2025.11179046.

## Selected Research Projects

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| <b>Exploring cooperation mechanisms via reinforcement learning in network common-pool resource games</b> | <b>2026.01 – Present</b> |
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*First Author | Manuscript in preparation*

- Established a networked common-pool resource games with resource allocation and cooperation dynamics.
- Developed a reinforcement learning-based agent responsible for allocation mechanisms to improve long-term resource sustainability and fairness.
- Used three baseline allocation mechanisms to fit and interpret the RL agent's policy, achieving better results.

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| <b>A tripartite intervention–conflict game with Frequency-Adjusted Q-learning</b> | <b>2025.09 – 2026.05</b> |
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*First Author | Applied Mathematical Modelling*

- Developed a tripartite evolutionary game model to analyze strategic interactions among interveners and conflicting parties.
- Established error bounds linking Frequency-Adjusted Q-learning and evolutionary game dynamics.
- Analyzed the effects of key parameters on system equilibria and stability.

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| <b>Recovering cooperation in spatial prisoner's dilemma via a forced loner mechanism</b> | <b>2025.11 – 2026.01</b> |
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*First Author | Physica A: Statistical Mechanics and its Applications*

- Proposed a resource-constraint dynamic model in spatial prisoner's dilemma games.

- Designed a forced loner mechanism inspired by social welfare safety nets to model resource-depletion-induced strategy transitions.
- Conducted numerical simulations to study cooperation level, resource sustainability, and spatial pattern formation.

## Design of transmodal associative learning circuit based on memristor 2023.12 – 2024.05

*Bachelor's Thesis | Harbin Institute of Technology (Shenzhen)*

- Designed a memristor-based associative memory circuit inspired by neuromorphic learning and cross-modal association.
- Constructed storage neuron, synapse, and retrieval neuron modules based on an AIST memristor model.
- Implemented bimodal and multimodal associative learning circuits and verified learning, forgetting, recovery, transfer, and consolidation behaviors through PSPICE simulations.

## Experience

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### ‘Zhuoyi Cup’ · 2025 Second National Swarm Intelligence Challenge 2025.08 – 2025.11

*Collaborator | First Prize in the Aerial Competition Category*

- Designed reinforcement learning-based strategies for multi-agent confrontation tasks.
- Built a rule-based expert system to collect demonstration data and used behavior cloning to warm-start policy learning.
- Contributed to code implementation, experimental evaluation, and the final competition presentation.

### Research on Reinforcement Learning with Role Assignment Across Scales 2025.12 – 2026.05

*Undergraduate Thesis Mentor | Undergraduate Thesis*

- Mentored an undergraduate thesis project on “Intelligent Game Modeling Based on Hierarchical Reinforcement Learning”.
- Guided the setup of the SMAC-based StarCraft II multi-agent environment, as well as experimental design, result analysis, and thesis writing.

### Nonlinear Systems Course Spring 2025 & Spring 2026

*Assistant | Shanghai Jiao Tong University*

- Served as a teaching assistant for the undergraduate course “Nonlinear Systems” over two spring semesters.
- Supported students by answering course-related questions with explaining key concepts, analytical methods, and problem-solving techniques.
- Responsible for grading homework assignments and exams.

## Technical Skills

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**Research Methods:** Evolutionary game theory, reinforcement learning, multi-agent systems, complex networks, numerical simulation.

**Programming and Tools:** Python, MATLAB, LaTeX, PyTorch, Git, Linux, PSPICE.

**Languages:** Mandarin Chinese (native), English (CET-6: 512; IELTS in preparation).

**Interests:** Reading, Music, Fitness, Hiking, Traveling, and Video games.